THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of $\sin^2 30^\circ + \sin^2 45^\circ$ is: 1

(a) $\frac{7}{4}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{2}$ **(d)** $\frac{3}{4}$

2. A boy whose eyes are 2 m above the ground stands 18 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) 20 m **(b)** 18 m **(c)** 21 m **(d)** $20\sqrt{3}$ m

3. The length of the tangent from an external point P to a circle is 16 cm and P is 34 cm from the centre. The radius of the circle is: 1

(a) 30 cm **(b)** 18 cm **(c)** 17 cm **(d)** 50 cm

4. A square is inscribed in a circle of radius 7 cm. The area of the circle outside the square is: 1

(a) 56 cm² **(b)** 105 cm² **(c)** 98 cm² **(d)** 154 cm²

5. If $\sin\theta = \frac{4}{5}$, then $\sec\theta + \tan\theta =$ 1

(a) $\frac{1}{3}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{9}{4}$ **(d)** 3

6. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 8 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) 4 m **(b)** $4\sqrt{3}$ m **(c)** $8\sqrt{3}$ m **(d)** $\frac{8\sqrt{3}}{3}$ m

7. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 30 cm. The length of each tangent is 16 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 136 cm **(b)** 32 cm **(c)** 46 cm **(d)** 92 cm

8. The largest possible circle is cut from a square sheet of side 56 cm. The area of the sheet left over is: 1

(a) 3136 cm² **(b)** 1344 cm² **(c)** 672 cm² **(d)** 2464 cm²

9. If $\cos\theta = \frac{3}{5}$, then $\csc\theta - \cot\theta =$ 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{1}{3}$ **(d)** 2

10. Two points on level ground lie on opposite sides of a tower of height $6\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. The distance between the two points is: 1

(a) 24 m **(b)** 48 m **(c)** $24\sqrt{3}$ m **(d)** 12 m

11. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 40 cm. The length of each tangent is 9 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 180 cm² **(b)** 98 cm² **(c)** 360 cm² **(d)** 720 cm²

12. A circular park of radius 20 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 1034 m² **(b)** 2068 m² **(c)** 517 m² **(d)** 880 m²

13. $\dfrac{\sin 20^\circ}{\cos 70^\circ} + \dfrac{\cos 30^\circ}{\sin 60^\circ} =$ 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 4

14. A storm breaks a tree at a height of 3 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The distance from the foot of the tree to the point where its top touches the ground is: 1

(a) 3 m **(b)** $3\sqrt{3}$ m **(c)** $\sqrt{3}$ m **(d)** 6 m

15. Two concentric circles have radii 52 cm and 20 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 96 cm **(b)** 64 cm **(c)** 72 cm **(d)** 48 cm

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. If $\tan A = \frac{4}{3}$, find $\dfrac{\sin A + \cos A}{\sin A - \cos A}$. 2

17. From the top of a building 6 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. Find the height of the tower. 2

18. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 30^\circ$. Find the $\angle APB$. 2

19. The minute hand of a clock is 28 cm long. Find the area swept by the minute hand in 15 minutes. 2

20. If $\sin(2x^\circ) = \frac{1}{2}$, find the value of x (an acute angle). 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. If $\sin(3x^\circ) = \frac{1}{2}$, find the value of x (an acute angle). 3

22. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 4 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 3

23. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (9x + 4)$ cm and $QB = (5x + 20)$ cm. Find the length of QA. 3

24. A circular field has radius 14 m. Fencing costs ₹ 12 per metre. Find the cost (in ₹) of fencing the field once. 3

OR

If $\tan\theta = \dfrac{15}{8}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

25. A building is $6\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

OR

A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 18$ cm, $BC = 15$ cm and $CD = 12$ cm. Find the length of AD. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Prove that: $(\sec A - \tan A)^2 = \dfrac{1 - \sin A}{1 + \sin A}$. 5

OR

Two vertical poles of equal height stand on opposite sides of a road 60 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

27. Tangents PA and PB are drawn from a point P at a distance of 52 cm from the centre O of a circle of radius 20 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Two concentric circles have radii 14 cm and 7 cm. A sector of angle 45° is marked in both. Find the area of the region between the two arcs within the sector. 5

END OF QUESTION PAPER Total: 50 Marks

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